

RESEARCH ARTICLE**A real-time reservoir inflow forecast evaluation framework****Cameron Bracken | Youngjun Son | Vince Tidwell | Nathalie Voisin**¹ Pacific Northwest National Laboratory,
Washington, USA**Correspondence**Corresponding author Cameron Bracken,
Email: cameron.bracken@pnnl.gov**Abstract**

New hydrologic forecast products give hydropower operators more options, but operators still need to determine whether a product improves decisions enough to justify its cost. We present a procedure for evaluating competing reservoir inflow forecasts as they are used. Operators help select and revise the metrics, and the evaluation uses the same data available in operations. Standard verification metrics are paired with measures tied to site-specific decisions. We apply the procedure to three reservoirs in the Great River Hydro system on the Connecticut River in the Northeast U.S. The commercial product labeled B generally matched the in-house forecast and outperformed product A. A cumulative-error stress test also showed that B had a lower probability of exceeding the forebay compliance band during the first day at all three dams. By days two and three, exceedance probabilities were high for both commercial products, showing the horizon at which passive use of either forecast would be untenable. The framework evaluates forecast quality rather than economic value; a value assessment requires a scheduling model, operating constraints, and market prices.

KEY WORDS

hydropower, streamflow forecasting, real-time evaluation, verification, validation

1 | INTRODUCTION

Environmental regulation, changing water demands, and operational uncertainty require reservoir and hydropower operators to make more precise decisions (e.g., Lee et al. (2025) and Guo and Liu (2024) and Badr et al. (2023) and Aljoda and Jain (2021)). This has increased interest in streamflow forecasting tools (Yang et al. 2021 and Pagano et al. 2014 and Liu et al. 2012). The practical question is whether a new product improves operations enough to justify its cost.

Hydrologic forecasting and hydropower scheduling have advanced through data driven models (Kratzert et al. 2019) and scheduling optimization (Zhang et al. 2024). Forecast verification still commonly relies on hindcasts and standard statistical metrics over long records. Hindcasts are often unavailable because they are costly to produce or require data that were not archived. Archived operational forecasts provide another option, but they do not capture products that change during the evaluation or forecasts adjusted by operators. We address the case in which a reservoir operator must compare products under current operating conditions without a hindcast.

No metric set fits every system. Meteorological and hydrologic data vary among basins, as do watershed properties and infrastructure (Pham et al. 2021). Operating rules range from run-of-river constraints to seasonal rule curves (Zajac et al. 2017). Decisions also change with lead time and flow regime. An evaluation must therefore connect statistical performance to the decisions made at a particular site.

Several programs compare forecasts under common conditions. HEPEX has organized community experiments and testbeds[†], and the U.S. Bureau of Reclamation has run forecast rodeos[‡]. Operational evaluations can fit more readily into existing practice than retrospective studies (Krajewski et al. 2021). However, a common ranking does not answer whether a product improves decisions at a specific utility enough to cover deployment, training, and transition costs.

We present an operating procedure for this setting. Operators participate in the experimental design, select and revise metrics, and interpret results against recent decisions. The evaluation uses the data available in daily operations. These features distinguish the procedure from a fixed retrospective verification.

We demonstrate the procedure at three Great River Hydro plants on the Connecticut River. The case study combines standard metrics with measures tied to day-ahead bidding and forebay compliance. The same process can be adapted to other systems, but the tailored metrics will differ.

2 | METHODS

While probabilistic and ensemble forecasts and evaluation metrics are gaining in popularity, deterministic forecasts such as those issued by the National Weather Service River Forecast Centers (RFC) are still widely used operationally and are adopted as a benchmark in this framework. All three products evaluated here are deterministic, so the metrics we describe are deterministic; Section 2.5 sets out how the framework extends to ensemble products, since the structure of the process does not depend on which kind of forecast is being evaluated. In addition, we focus on metrics which are feasible to compute in a real-time setting as opposed to a hindcast based evaluation.

Hindcasts remain the strongest basis for evaluating forecast quality because they provide large samples and a broad range of hydrologic conditions. Comparing hindcast inflow with observed or calculated inflow does not require reconstructing operations. Reconstructing forecasts that include human judgment is harder because the forecaster would need the information available at each historical issue time. Historical day-ahead and real-time wholesale prices can often be retrieved, but outages, constraints, operator judgment, and the way price signals affected decisions may not be recoverable. Extending the analysis from quality to value also requires a reservoir operations model or operator input. A real-time evaluation builds the forecast archive while these conditions occur.

[†] www.hepex.org

[‡] <https://www.usbr.gov/research/challenges/forecastrodeo.html>

2.1 | Real-time inflow forecast evaluation

A general real-time forecast evaluation process can be described in the following steps. Steps 1 through 3 are generic practice, common to any verification exercise. Steps 4 through 6 are where this framework departs from retrospective verification.

1. **Collect real-time data** - This may include streamflow forecasts (natural or regulated), calculated inflows, scheduled and actual outflows, and other data relevant for computing evaluation-relevant metrics such as forecast issue times and past forecasts.
2. **Compute evaluation metrics** - At the start of an evaluation, it may only be possible to compute a limited number of generic evaluation metrics but as the available data grows it may become possible to compute more reliable metrics as well as metrics tailored to the specific system.
3. **Visualize results** - Use figures that allow operators to compare products without extensive explanation.
4. **Gather feedback** - Collecting feedback on an ongoing basis from operators is critical to the success of a real-time forecast evaluation. Operators should be asked which recent forecast errors mattered operationally and which did not, since the answers are not always apparent from the metrics alone.
5. **Update metrics** - Feedback is used to add metrics tailored to the system or reservoir being evaluated, and to remove or modify metrics that have proven uninformative. The set of metrics is therefore not fixed at the outset but develops as the evaluation proceeds.
6. **Iterate** - The above steps should be repeated at a regular interval (monthly or weekly) throughout the entire forecast evaluation period, ideally at least one year in duration.

The mechanics of steps 4 and 5 determine what the evaluation produces, so an example of how feedback was collected and acted on in our case helps clarify this point. Feedback was collected in recurring meetings with the operators, held monthly for most of the evaluation. Each meeting reviewed the current calculated metrics against recent operating experience, and operators were asked to identify which recent forecast errors mattered operationally, and which did not. The answers frequently diverged from what the generic metrics indicated, especially during the high-flow season. Operators discounted large errors during high-flow spill conditions, when they had little discretion over releases and forecast quality was operationally irrelevant, while treating moderate errors with the potential to impact the forebay compliance band as serious. These ongoing discussions motivated the tailored metrics in Section 3.6: the forebay exceedance metrics were designed in direct response to operators stating that compliance risk, rather than average error, determined whether they trusted a forecast.

Metric development and data collection occur on different timescales. In this case the metric set stabilized within about six months, while the evaluation continued to sample more conditions. For applications with strong seasonality, including this

one, we recommend at least one annual cycle because product rankings can vary by season (Section 3.5). A shorter evaluation remains conditional on the seasons and events observed.

2.2 | Benchmark forecasts

Benchmark forecasts put product performance in context. A “perfect” forecast substitutes observed or calculated inflow for a forecast and provides an upper reference. A persistence forecast carries the most recent observation through the forecast horizon. Persistence can be competitive from hours to about two days, but it degrades quickly. Climatology is usually a more suitable lower reference at weekly to seasonal horizons. The benchmark should match the decision horizon and may also include the forecast currently used at the reservoir.

NSE and KGE also depend on the distribution of observations at each location, so raw values require care in cross-site comparisons (Knoben et al. 2019). NSE measures improvement over the observed mean. A forecast can therefore have a higher NSE at a more variable site even when its absolute errors are larger. We report RMSE and reference forecasts alongside NSE and KGE so readers can see both absolute error and performance relative to simple alternatives.

2.3 | Smoothing calculated inflows

Reservoir inflow is often calculated from changes in forebay elevation and an elevation-volume relationship rather than measured directly. Short-term changes caused by wind, turbine and spillway releases, seiches, and upstream waves can produce noise and nonphysical values. Smoothing may therefore be appropriate before comparing calculated and forecast inflow. We used two successive moving averages, which produce a triangular kernel, but other filters are available (Elshorbagy et al. 2002). The window must be reported and tested because smoothing changes the evaluation target.

2.4 | Traditional inflow evaluation metrics

Deterministic inflow forecasts can be evaluated using measures of average error, correlation, and bias. Higher moments, categorical performance, and conditional statistics may also be useful (St-Aubin and Agard 2022 and Hewamalage et al. 2023). We use a small set of metrics that can be expanded for a particular application. Each should be calculated by lead time and season because aggregate values can hide changes in performance.

Average hydrologic forecast behavior is typically measured with an error metric such as root mean squared error (RMSE), Nash-Sutcliffe Efficiency (NSE) (Nash and Sutcliffe 1970), or Kling-Gupta Efficiency (KGE) (Gupta et al. 2009). We recommend KGE because it decomposes forecast performance into correlation, a variability ratio, and a bias ratio, so that a low score can be

attributed to a specific deficiency rather than only registered as large aggregate error. NSE and RMSE aggregate squared error into a single number, which makes them more strongly influenced by the largest errors and less diagnostic about their source. The KGE is given by

$$\text{KGE} = 1 - \sqrt{(r - 1)^2 + \left(\frac{\sigma_f}{\sigma_o} - 1\right)^2 + \left(\frac{\mu_f}{\mu_o} - 1\right)^2} \quad (1)$$

where r is the linear correlation coefficient, σ_f and σ_o are forecast and observed standard deviations, and μ_f and μ_o are forecast and observed means. KGE values greater than -0.41 improve on the observed mean as a predictor (Knoben et al. 2019); the maximum is 1. NSE and KGE have different properties and should not be used interchangeably. We report both and use RMSE, bias, correlation, and benchmark forecasts to interpret them.

The error, or residual, of a single forecast is the difference between the forecast and the observation. Bias is the systematic component of error, the tendency of a forecast to be too high or too low on average, and is conventionally expressed for a set of forecasts as a percentage of total observed volume

$$\text{PBIAS} = 100 [\%] \times \frac{\sum(f_i - o_i)}{\sum(o_i)} \quad (2)$$

where positive values indicate forecasts greater than observations and negative values indicate forecasts less than observations.

Linear association between observations and forecasts is measured using the Pearson correlation coefficient

$$r = \frac{\sum(o_i - \mu_o)(f_i - \mu_f)}{\sqrt{\sum(o_i - \mu_o)^2 \sum(f_i - \mu_f)^2}} \quad (3)$$

where r can range from -1 to 1 but for forecast evaluation, negative values would indicate poor forecast performance.

Metrics should be computed on a common sample when products cover different periods. Skill scores relative to a reference forecast are also useful, but the evaluated and reference forecasts must use the same observations. Otherwise differences in sample variance can be mistaken for differences in skill.

2.5 | Extension to probabilistic and ensemble forecasts

The process applies to deterministic and probabilistic forecasts; the metrics and figures change. Probabilistic products can describe risk around compliance or market decisions. The continuous ranked probability score (CRPS), rank histograms, and reliability diagrams are common choices. CRPS reduces to mean absolute error for a deterministic forecast, which permits comparison on a common scale.

3 | CASE STUDY: GREAT RIVER HYDRO

Great River Hydro (GRH) is New England's largest conventional hydropower generator. As a private, independent power producer, GRH owns and operates 589 megawatts of installed nameplate hydropower capacity across 13 generating stations on the Connecticut and Deerfield Rivers in ISO New England. In this evaluation we focused on the inflow to three main hydropower producing dams in the system, Wilder, Bellows Falls, and Vernon (Figure 1) as well as three main tributaries, the Ottauquechee, Sugar and White Rivers.

The three dams lie in series on the main stem of the Connecticut River, with Wilder upstream, followed by Bellows Falls, then Vernon. The three projects are operated jointly, so releases from each become inflow to the next. The only significant local flows occur between Wilder and Bellows Falls in the form of the three tributaries mentioned above. These three sites were selected from the larger set shown in Figure 1 because they are the largest generators on the system and they are where the operators concentrate their scheduling attention. In addition, these projects are part of an ongoing relicensing process through the Federal Energy Regulatory Commission (FERC), which includes new limits to forebay fluctuations. The remaining facilities in Figure 1 are smaller and are not covered by the same license conditions.

Basin hydrology is snowmelt influenced, with a spring freshet typically spanning March through May, convective rainfall in summer, and rain and rain-on-snow driven events in winter. Several tributaries have upstream flood control regulation by the U.S. Army Corps of Engineers, which contributes to inflow variability that is not directly weather driven.

TABLE 1 Physical characteristics of the six evaluation points. Drainage areas are from the USGS National Water Information System for the reference gage listed. Mainstem areas are nested, so the area above Vernon includes those above Bellows Falls and Wilder.

Location	Type	USGS reference gage	Area (mi ²)	Area (km ²)
Wilder	Mainstem dam	01144500 West Lebanon, NH	4,092	10,598
Bellows Falls	Mainstem dam	01154500 North Walpole, NH	5,493	14,227
Vernon	Mainstem dam	01156500 Vernon, VT	6,266	16,229
White River	Tributary	01144000 West Hartford, VT	690	1,787
Sugar River	Tributary	01152500 West Claremont, NH	269	697
Ottawuechee River	Tributary	01151500 North Hartland, VT	221	572

3.1 | Operating context

The three dams have limited storage relative to their inflow and are operated as run-of-river facilities. Their daily cycle consists of an upward ramp, a period of flexible generation, a downward ramp, and a refill period. Generation is bid into the ISO New England day-ahead market, so the day-ahead inflow volume is the quantity that matters most to the operators, since it sets what they can bid into the market. The binding operational constraint comes from the FERC relicensing of the three lower Connecticut River dams, which imposed a ± 0.5 foot (≈ 15 cm) limit on forebay fluctuation, with penalties for exceedance. The central

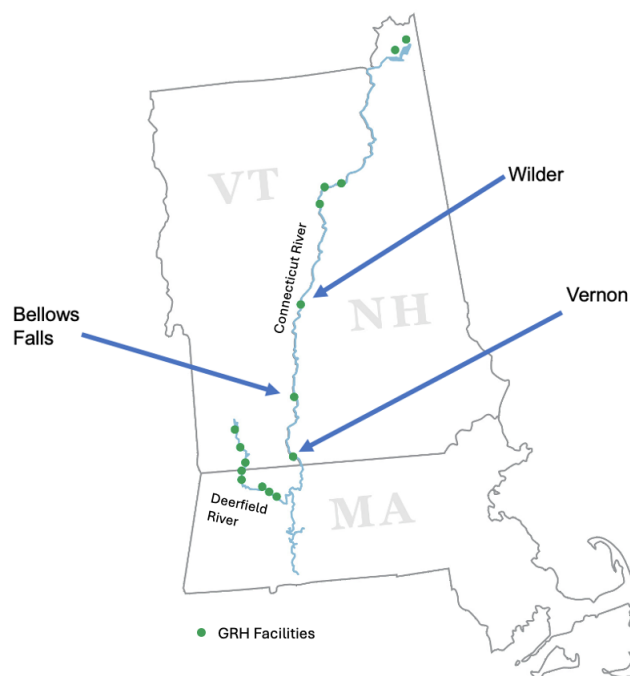


FIGURE 1 The Great River Hydro system in the Northeast U.S. with the locations of the three major dams in this evaluation.

operating problem is therefore committing generation based on an uncertain inflow forecast without violating the forebay limits. This is what motivated the cumulative-error and threshold-exceedance metrics in Section 3.6.

At the time of the evaluation the operators used an in-house forecast built from National Weather Service River Forecast Center forecasts, routed to the dam locations and adjusted using operator judgment. This is the incumbent product against which the two commercial forecasts were evaluated, and it is labeled GRH in the results.

Setting up a real-time evaluation required the development of an automated data platform that enabled sharing of real-time system data as well as the forecasts. This was necessary to share certain data with external parties for the purposes of forecast evaluation. While such a system may not be necessary for evaluations conducted internally by a single organization, some form of data access (API, database, etc.) is required for a real-time forecast evaluation. Such systems incur an up-front cost that should be considered when deciding on an evaluation approach. In addition, real-time evaluations come with all of the challenges of any operational system, such as system uptime, latency, and maintenance. These factors may help decide if a forecast is viable before any evaluation is done; a forecast that cannot be produced operationally should not be considered in a real-time evaluation.

We ran the real-time evaluation for almost two years, from June 2023 to April 2025, with regular feedback from the operators. We evaluated two commercial forecast products which are anonymized and labeled as A and B. In addition we evaluated the GRH in-house forecast which is a combination of RFC forecasts, routing, and operator judgment. During the initial few months of the evaluation we focused on generic metrics of forecast performance. As the quantity of data increased we transitioned to seasonally computed metrics and tailored metrics developed in collaboration with the system operators.

The three products differ in the coverage they provided over this period, which constrains how they can be compared. Forecast B was available for the full period, giving 657 day-ahead forecasts at each dam. Forecast A was discontinued in November 2024 and the in-house forecast archive also ends in November 2024, giving 539 and 528 day-ahead forecasts respectively. The three products overlap on 510 days at each dam, from 3 June 2023 to 20 November 2024. Metrics computed over each product's full available record, which is what we report, are therefore not computed over identical samples. We repeated the calculation restricted to the 510 common days and the ranking of the three products is unchanged at all three dams, with individual metrics shifting by no more than 0.03 NSE, so we report the full-record values so as not to discard any data. This kind of partial coverage reflects the operational realities of any real-time evaluation.

3.2 | Forecast products

Anonymization of the commercial products limits reproducibility, so we describe each to the extent our agreements allow. Forecast A is produced by a consulting firm using a conceptual rainfall-runoff model with hydraulic routing through the Connecticut River main stem, driven by a commercial numerical weather prediction forecast, and issued once daily at an hourly time step out to roughly 8 days. Forecast B is produced by a technology firm using a machine learning model trained on gauge and remotely sensed catchment observations, issued twice daily at an hourly time step out to 10 days. The GRH in-house forecast combines RFC forecasts with routing and operator judgment and extends only through the day-ahead period, roughly 32 hours, because that is the horizon it is used for. Neither vendor released model internals or calibration and validation diagnostics, and we did not have access to them. A utility evaluating commercial forecast products generally cannot inspect them and must judge them on operational performance, which is part of the motivation for this framework. A full procurement evaluation should also weigh the operational cost, data dependencies, and ease of integration of each product alongside its skill, since these differ substantially between the two commercial products evaluated here.

3.3 | Hydrology of the evaluation period

The evaluation period included the July 2023 Vermont flood. This event produced the largest calculated inflows of the evaluation at Vernon (83,100 cfs, 2,353 m³/s) and Bellows Falls (67,000 cfs). A December 2023 rain-on-snow event produced the largest inflow at Wilder (27,600 cfs) and the second largest at Vernon and Bellows Falls. The evaluation period also included spring freshets of 2024 and 2025 and dry summer conditions in both 2023 and 2024. The period therefore sampled a reasonable range of the conditions relevant to GRH operations. It did not, however, sample the full distribution of hydrologic conditions. Notably, the July 2023 event produced one of the ten largest flows ever recorded on the main stem of the Connecticut River. The evaluation

also did not include any severe multi-year drought. Any real-time evaluation is bounded in this way, and the conclusions below should be taken as conditional on the observed period.

Most weather-driven inflow forecasts extend 7–10 days, but GRH relies most on next-day volume because it informs the day-ahead market bid. Table 2 compares the two commercial products, the in-house forecast, and the perfect and persistence benchmarks for this horizon. Over the evaluation period, forecast B and the in-house forecast performed similarly and both outperformed A. Lead-time and monthly results below show variation hidden by the aggregate table.

Table 2 also shows why cross-location comparisons need more than one metric. Forecast A has its highest NSE at Vernon (0.81) and its lowest at Bellows Falls (0.55), yet its RMSE is 5,097 cfs at Vernon, 5,904 cfs at Bellows Falls, and 2,278 cfs at Wilder. NSE is normalized by observed variance, so a higher value at one site does not imply a smaller error than at another. The persistence benchmark also changes sharply among sites, from an NSE of -0.81 at Wilder to 0.80 at Vernon. We therefore use the table to compare products within a location, while RMSE and the benchmark rows provide context across locations. Metrics computed on the 510 days common to all products preserve the within-location ranking.

TABLE 2 Day-ahead forecast evaluation metrics for June 2023 to April 2025. A and B are commercial products, GRH is the in-house forecast, and perfect and persistence are benchmarks. The products cover different portions of the period; n is the number of forecasts evaluated. Values should be compared primarily within a location because NSE and KGE depend on the local observation distribution.

Location	Forecast	n	NSE	KGE	PBIAS (%)	RMSE (cfs)	r
Wilder	A	539	0.72	0.83	-8.0	2,278	0.88
	B	657	0.83	0.88	0.1	1,786	0.93
	GRH	528	0.88	0.89	-9.9	1,476	0.95
	perfect	657	1.00	1.00	0.0	0	1.00
	persistence	657	-0.81	0.02	-75.9	5,866	0.75
Bellows Falls	A	539	0.55	0.77	0.4	5,904	0.77
	B	657	0.91	0.95	0.6	2,615	0.95
	GRH	528	0.85	0.88	-10.4	3,309	0.93
	perfect	657	1.00	1.00	0.0	0	1.00
	persistence	657	0.39	0.35	-39.2	6,831	0.87
Vernon	A	539	0.81	0.83	-11.3	5,097	0.91
	B	657	0.89	0.94	1.7	3,872	0.94
	GRH	528	0.89	0.88	-11.0	3,764	0.95
	perfect	657	1.00	1.00	0.0	0	1.00
	persistence	657	0.80	0.69	-18.1	5,167	0.93

Figure 2 shows the performance of all the available forecasts and benchmarks out to a 10 day lead time. The GRH in-house forecast is only used to develop bids for the day-ahead market so it does not extend the full 10 day period. At Wilder and Vernon all three forecasts perform roughly similar with 1-3 days of lead time with the A and B forecasts diverging after that period. At Bellows Falls the three forecasts are more distinct in terms of quality with the B showing the best performance at most lead times followed by the GRH in-house forecast and then forecast A. Both commercial forecasts develop a growing negative bias with lead time, and the bias grows further downstream, consistent with errors propagating through the cascading reservoirs. For forecast A, day-ahead PBIAS is 0.4% at Bellows Falls and grows to -19.1% by day 7; for forecast B the corresponding values

are -1.3% and -7.8% . Because releases from each dam become inflow to the next, an error propagates directly to the next downstream project, and the effect compounds with lead time.

3.4 | Smoothing GRH calculated inflows

Figure 3 shows calculated inflow before and after smoothing. We applied two successive 24-hour moving averages. Their convolution yields a triangular kernel spanning 47 hourly observations; “24-hour” refers to the width of each component moving average. We selected this width to match the day-ahead volumetric decision. We repeated the evaluation with component windows of 0, 6, 12, and 24 hours (Table 3). The largest NSE change was 0.05, and the product ranking was unchanged at every location. Forecast B led A at Bellows Falls by 0.39 NSE without smoothing and by 0.37 with the 24-hour component window. Evaluators should report the filter and test the sensitivity of conclusions to its width.

TABLE 3 Sensitivity of day-ahead NSE to the component moving-average width used to smooth calculated inflow. Each nonzero width is applied twice, producing a triangular kernel. The 24-hour column corresponds to Table 2. Product rankings are unchanged.

Location	Forecast	0 h	6 h	12 h	24 h
Wilder	A	0.70	0.72	0.72	0.72
	B	0.82	0.83	0.83	0.83
	GRH	0.87	0.87	0.88	0.88
Bellows Falls	A	0.50	0.51	0.52	0.55
	B	0.90	0.90	0.90	0.91
	GRH	0.84	0.83	0.84	0.85
Vernon	A	0.79	0.79	0.79	0.81
	B	0.88	0.88	0.88	0.89
	GRH	0.88	0.88	0.88	0.89

3.5 | Seasonal performance

Figure 4 shows systematic monthly variation in forecast quality. Skill is highest during the spring snowmelt period, roughly March through May, for all three products. It is lowest in late summer and fall for the commercial products. Forecast A also degrades sharply in January and February, when day-ahead KGE pooled across the three dams falls to 0.45 and -0.67 , respectively; the February value is below the plotted range. Performance is poorest when inflow depends on rain, rain-on-snow, or convective storms. These events depend on uncertain precipitation forecasts. Snowmelt inflow is more predictable because the snowpack can be observed before it melts and temperature forecasts are more reliable than precipitation forecasts. Forecast A outperforms B in April and May, so no product leads in every month.

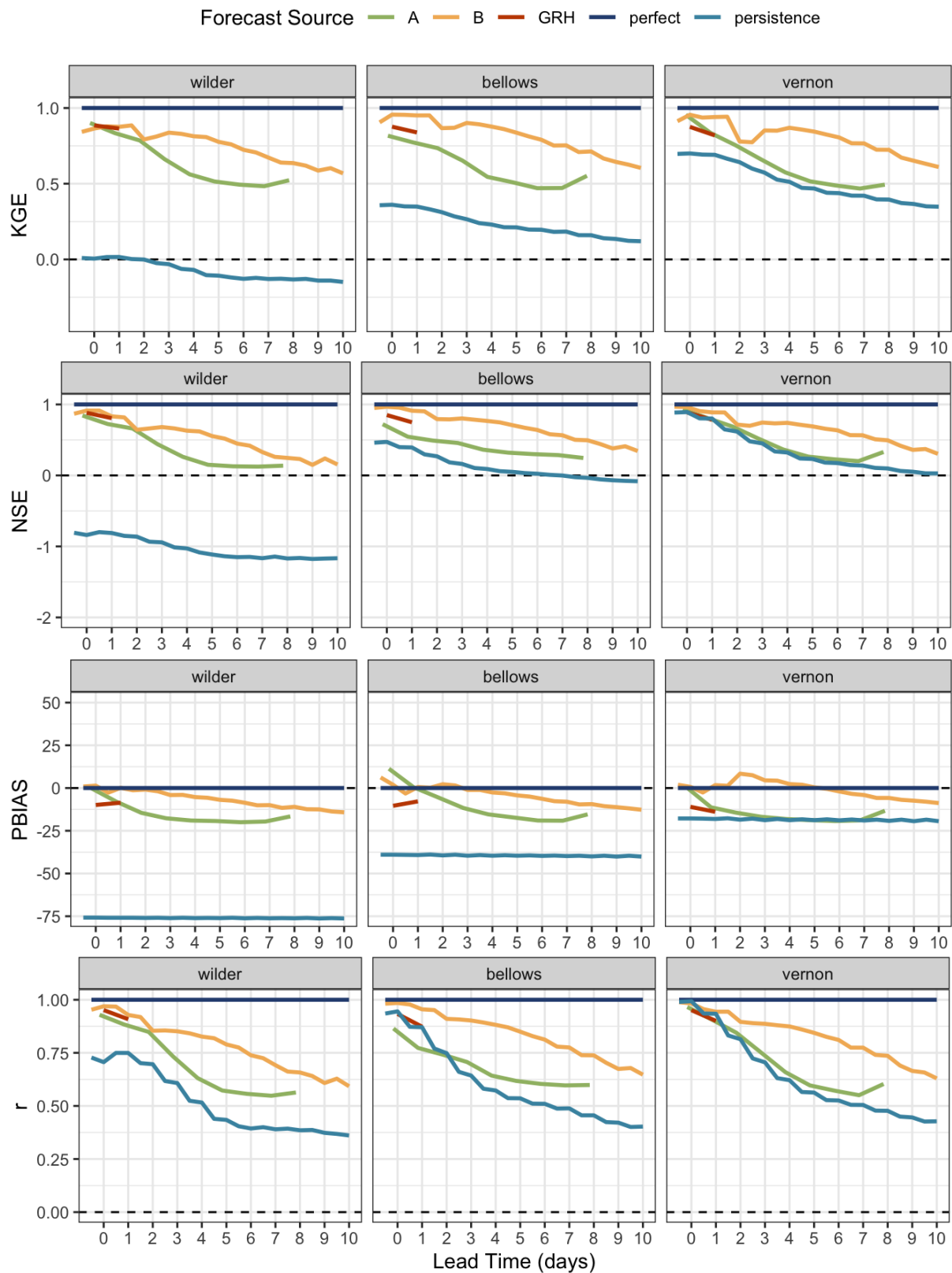


FIGURE 2 Forecast performance metrics by lead time. A and B are commercial forecasts, GRH is an in-house forecast. The dashed line indicates the threshold where a forecast has no skill over the mean of the observations, which is NSE of 0 and KGE of -0.41 . The persistence benchmark is a meaningful reference only over the first two days; beyond that it is shown for reference purposes only.

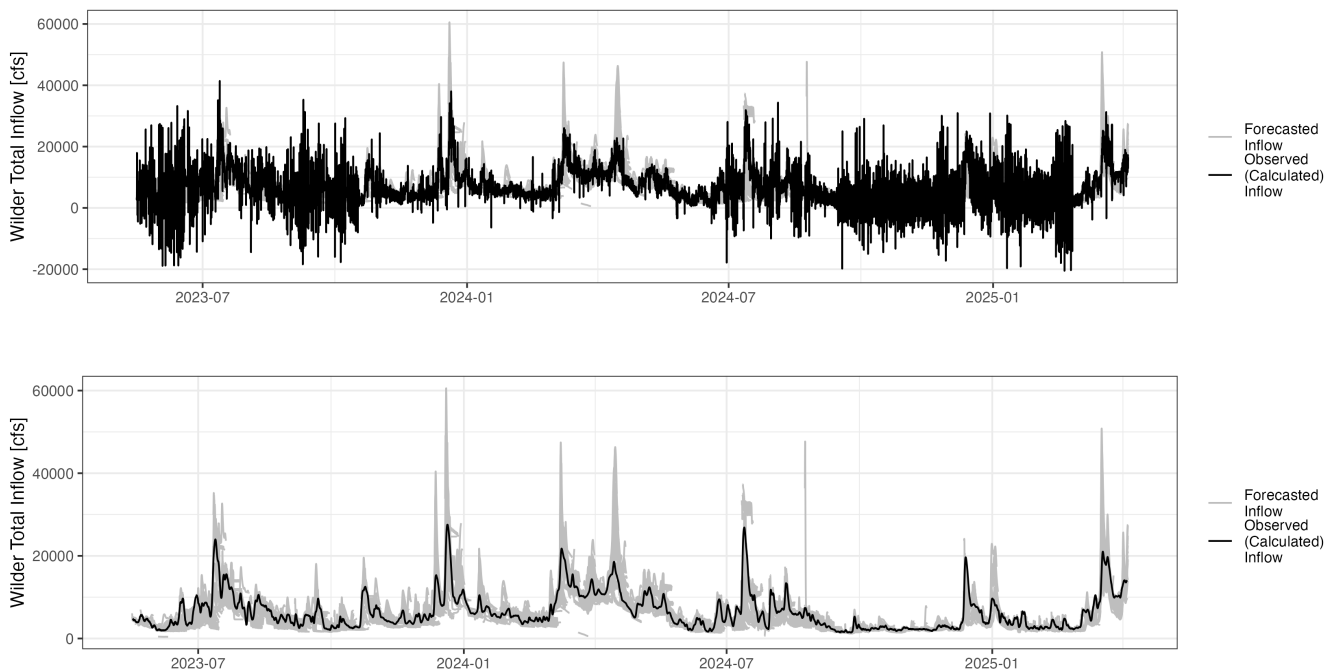


FIGURE 3 Unsmoothed (top) and smoothed (bottom) calculated inflow at Wilder Dam. Black lines show calculated inflow derived from forebay observations and dam outflow; gray lines show forecasts. Smoothing uses two successive 24-hour moving averages, yielding a triangular kernel spanning 47 hourly observations. Flows are in cubic feet per second (1,000 cfs $\approx$ 28.3 m³/s).

3.6 | Tailored metrics

In addition to the generic forecast evaluation metrics described above, we developed several tailored metrics specifically relevant to the GRH system and its operations. Each arose from the feedback process described in steps 4 and 5 of Section 2.1, and Table 4 traces each metric to the operator concern that prompted it. The first focused on day-ahead forecast performance as shown in the previous section. This was motivated by the day-ahead market that GRH bids into, as such they focus their attention heavily on the 1-2 day ahead period where the forecasts matter the most. Operationally, they adjust their flows daily, so no forecast is used after the 2 day mark.

TABLE 4 Tailored metrics developed during the evaluation, the operator concern that prompted each, and the approximate point in the evaluation at which it was added. Metric development converged within roughly the first six months.

Metric	Operator concern	Added
Day-ahead volumetric skill	Day-ahead market bids are set from next-day inflow volume, not hourly shape	Months 1–2
Seasonal metric breakdown	Products were suspected to behave differently in snowmelt and rain driven regimes	Months 3–4
Cumulative forebay error	Average error does not indicate whether the forebay band would be breached	Months 4–6
Forebay exceedance probability	Operators needed to know how quickly cumulative forecast error could exhaust the forebay band	Months 5–6

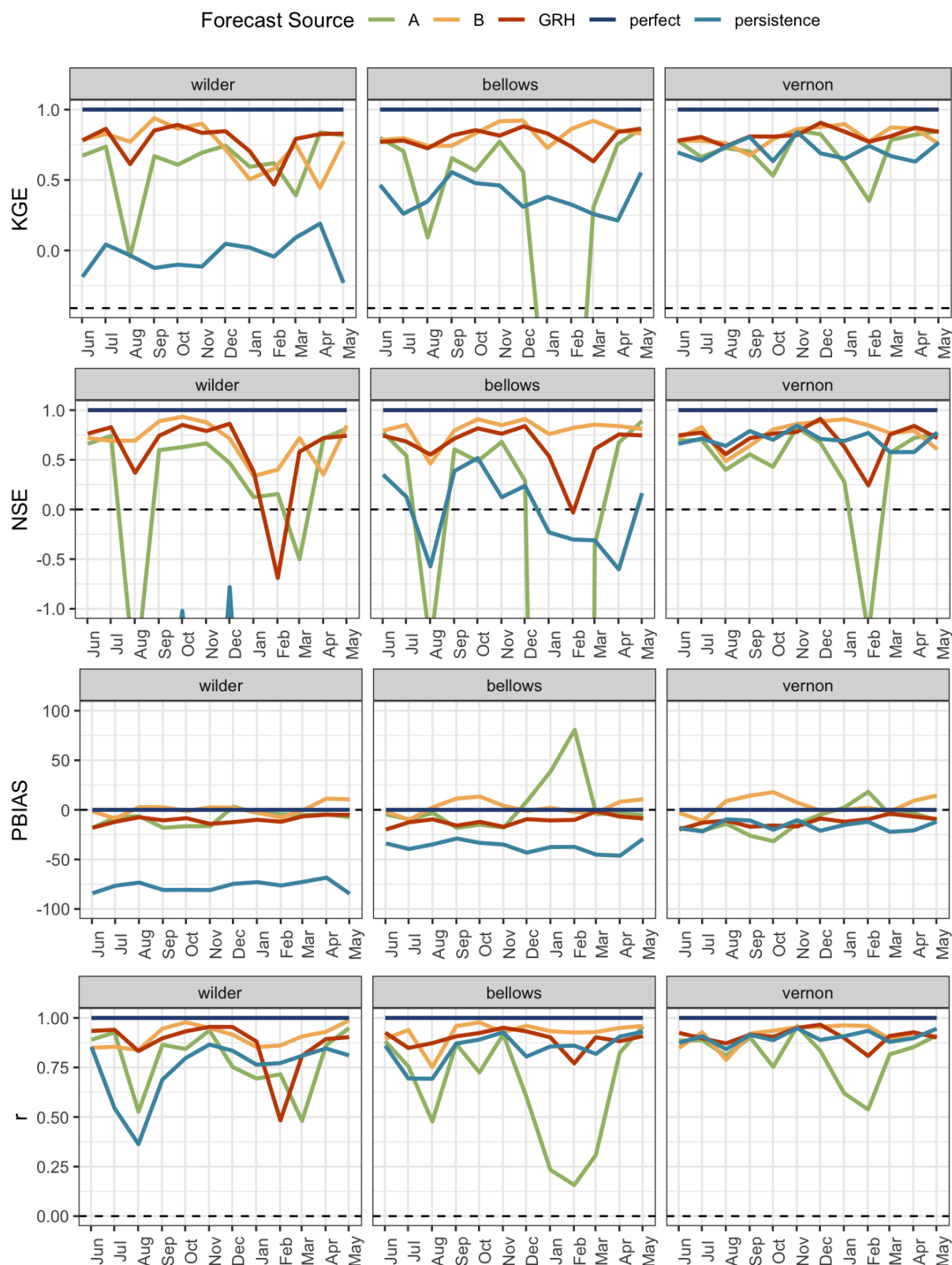


FIGURE 4 Forecast performance metrics by month. A and B are commercial forecasts, GRH is an in-house forecast. The dashed line indicates the threshold where a forecast has no skill over the mean of the observations, which is NSE of 0 and KGE of -0.41.

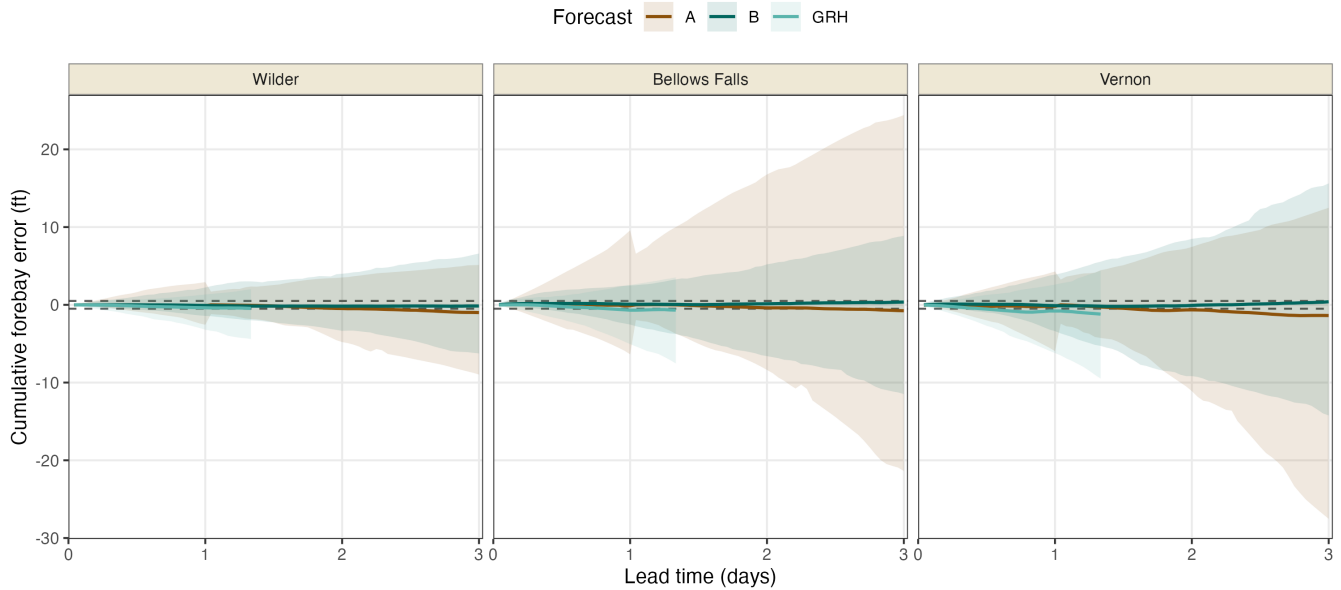


FIGURE 5 Cumulative forebay error under an idealized pass-inflow stress test with a neutral starting position. Lines show the median across forecast issues and shaded bands span the 2.5th to 97.5th percentiles. Dashed lines mark the ± 0.5 foot (≈ 15 cm) forebay operating band.

The FERC relicensing described in Section 3.1 imposed a ± 0.5 foot (≈ 15 cm) forebay operating band. We calculated cumulative hourly forecast error within each forecast issue and converted the accumulated volume to an equivalent forebay elevation change using an approximate storage scale of 30,000 cfs-hours per foot. Figure 5 shows the median and middle 95% of the resulting trajectories. This is an idealized stress test: it assumes forecast inflow is passed directly as outflow from a neutral forebay position. GRH does not operate this way. Operators schedule generation to market bids, depart from inflow within the day, and adjust as new information arrives. The metric is therefore neither a prediction nor a bound on observed compliance rates. Estimating realized risk requires a scheduling model and operator review.

Figure 6 shows the corresponding probability of leaving the forebay band. During the first 24 hours, forecast B has lower exceedance probability than A at all three dams: 30.8% versus 50.0% at Wilder, 47.5% versus 73.0% at Bellows Falls, and 53.7% versus 76.3% at Vernon. The in-house forecast is available only through the day-ahead period and has probabilities of 53.9%, 67.6%, and 73.2%, respectively. By 48 hours, the two commercial products converge to 58.1–76.2% across the three dams; by 72 hours, probabilities range from 73.1% to 90.3%. Thus B provides a clearer advantage over the first day, while neither commercial product can be passed through without intervention over longer horizons. This distinction was not apparent from the aggregate metrics alone.

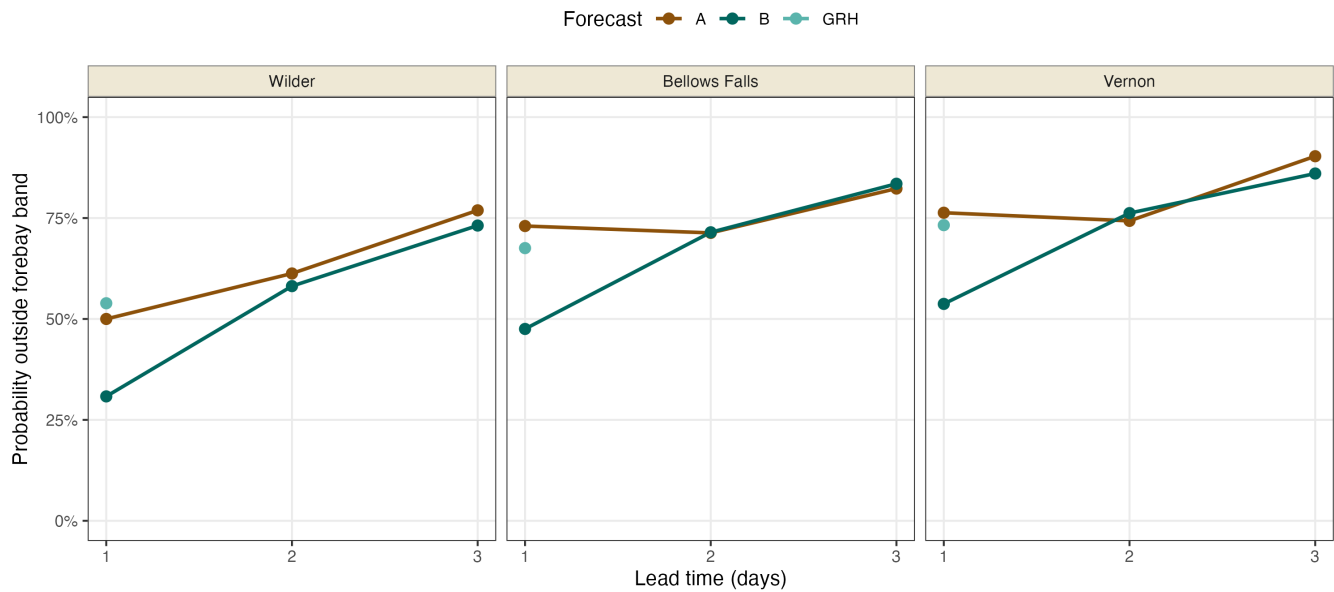


FIGURE 6 Probability of exceeding the ± 0.5 foot (≈ 15 cm) forebay operating band at daily lead times under the idealized pass-inflow stress test. Each forecast issue begins from a neutral forebay position. The in-house forecast ends after the day-ahead period.

4 | DISCUSSION

We presented a procedure for comparing reservoir inflow forecasts under current operating conditions when hindcasts are unavailable. It combines standard verification with metrics developed from operator concerns. At GRH, product B generally matched the in-house forecast and outperformed A. The forebay stress test identified a day-ahead advantage for B that aggregate metrics did not express in operational terms. The procedure supports product screening, but it does not by itself estimate revenue or compliance outcomes.

4.1 | Comparison with retrospective forecast evaluation

A retrospective evaluation can provide larger samples, more extremes, and broader hydrologic coverage than the less than two years available here. Forecast quality can be evaluated directly from hindcasts. Reconstructing value is more difficult. Historical wholesale prices may be available, but outages, constraints, operator judgment, and the response to market conditions may not be. A real-time evaluation observes these conditions as they occur, but it cannot replace a hindcast where one is feasible. The two approaches answer complementary questions.

4.2 | From forecast quality to forecast value

The metrics presented here measure forecast quality, not forecast value. Quality describes the relation between forecasts and observations. Value also depends on the decision, its consequences, and the decision maker's risk preferences. A more accurate forecast has no added value when it does not change an action. Classical cost-loss analysis links forecast use to decision utility (Thompson and Brier 1955). The relative utility value framework extends this logic to binary, multicategory, and continuous decisions and can represent risk aversion (Laugesen et al. 2023). Farkas et al. (2025) review related approaches.

The tailored metrics move toward value by expressing error in units tied to a decision. The forebay metric is still a stress test, not a valuation, because it omits actual schedules and operator responses. A GRH value study would first screen products using the quality evaluation, then route the archived forecasts through a scheduling and dispatch model containing reservoir constraints, unit performance curves, day-ahead and real-time prices, and the incumbent operating strategy. Benefits should be measured relative to the forecast and decisions currently used, with sensitivity tests for risk preferences and alternative operating strategies.

4.3 | Limitations and next steps

Forecast B performed better overall, but it did not lead in every month or at every horizon. Product selection should therefore consider the decisions and seasons that matter most. Probabilistic forecasts can describe uncertainty, although operators must still submit fixed market bids.

A second lesson concerns comparisons among locations. Raw efficiency metrics depend on observed variability, so they should not be used alone to rank sites. RMSE, sample coverage, and reference forecasts provide necessary context. Comparisons among products should use the same dates whenever coverage differs.

We did not estimate how forecast improvements affect operations or revenue. That analysis would couple the archived forecasts to a hydropower scheduling model. A model configured to reproduce the current bidding strategy may understate the benefit of a more accurate forecast because the strategy was developed around the incumbent product. A value study may therefore need to revise the scheduling model and operating strategy along with the forecast input.

The record length limits the real-time evaluation. The period included a regional flood and a rain-on-snow event (Section 3.3), but no multi-year drought. We could not assess product performance under sustained water-limited conditions. A multi-decade hindcast would cover more hydrologic regimes.

The choice between real-time and retrospective evaluation depends on the system and available resources. We recommend both when feasible. For this seasonally varying system, at least one year of real-time evaluation is appropriate. A shorter period samples only part of the annual cycle, so its conclusions are conditional on the seasons and events observed.

The reference inflows are calculated from forebay elevation changes and dam outflow. The observations therefore depend on operations and are not independent in the way a gauge record would be. The three products also cover different portions of the evaluation period (Section 3.1), so Table 2 uses overlapping but nonidentical samples. Restricting the analysis to common dates does not change the ranking. We report the full records to retain the available data.

Operator feedback determines which errors matter and which metrics should be revised or removed. Generic metrics support comparison. Tailored metrics connect errors to site constraints. Together they provide the forecast archive and quality screen needed for a later scheduling and value analysis.

DATA AND CODE

Code for the evaluation is freely available [§]. Sample data can be accessed from the data repository on Zenodo[¶].

Note that a pre-print non-peer reviewed draft of the manuscript has been uploaded to <https://eartharxiv.org/repository/view/10530/>.

AUTHOR CONTRIBUTIONS

Cameron Bracken - Conceptualization, Data Curation, Formal Analysis, Investigation, Methodology, Software, Validation, Visualization, Writing - Original Draft Preparation. Youngjun Son - Conceptualization, Data Curation, Formal Analysis, Investigation, Software, Validation, Visualization, Writing - Original Draft Preparation. Vince Tidwell - Conceptualization, Formal Analysis, Investigation, Methodology, Project Administration, Supervision, Writing - Original Draft Preparation. Nathalie Voisin - Conceptualization, Methodology, Project Administration, Supervision, Funding Acquisition, Writing - Original Draft.

CONFLICT OF INTEREST

The authors declare no potential conflict of interests.

REFERENCES

- Aljoda, Ali, and Shaleen Jain. "Uncertainties and Risks in Reservoir Operations under Changing Hydroclimatic Conditions." *Journal of Water and Climate Change*, vol. 12, no. 5 August 2021), pp. 1708–1723.
- Badr, Ahmed, et al. "Dam System and Reservoir Operational Safety: A Meta-Research." *Water*, vol. 15, no. 19 January 2023), pp. 3427.
- Elshorbagy, A., et al. "Noise Reduction in Chaotic Hydrologic Time Series: Facts and Doubts." *Journal of Hydrology*, vol. 256, no. 3 January 2002), pp. 147–165.
- Farkas, Hannah, et al. *The Economic Value of Weather Forecasts: A Quantitative Systematic Literature Review*, Washington, DC: World Bank, September 2025.
- Guo, Yuxue, and Li Liu. "Reservoir Control Operations and Water Resources Management." *Water*, vol. 16, no. 20 January 2024), pp. 3000.
- Gupta, Hoshin V., et al. "Decomposition of the Mean Squared Error and NSE Performance Criteria: Implications for Improving Hydrological Modelling." *Journal of Hydrology*, vol. 377, no. 1–2 October 2009), pp. 80–91.
- Hewamalage, Hansika, et al. "Forecast Evaluation for Data Scientists: Common Pitfalls and Best Practices." *Data Mining and Knowledge Discovery*, vol. 37, no. 2, 2023, pp. 788–832.
- Knoben, Wouter J. M., et al. "Technical Note: Inherent Benchmark or Not? Comparing Nash–Sutcliffe and Kling–Gupta Efficiency Scores." *Hydrology and Earth System Sciences*, vol. 23, no. 10 October 2019), pp. 4323–4331.
- Krajewski, Witold F., et al. "Real-Time Streamflow Forecasting: AI vs. Hydrologic Insights." *Journal of Hydrology X*, vol. 13 December 2021), pp. 100110.
- Kratzert, Frederik, et al. "Towards Learning Universal, Regional, and Local Hydrological Behaviors via Machine Learning Applied to Large-Sample Datasets." *Hydrology and Earth System Sciences*, vol. 23, no. 12 December 2019), pp. 5089–5110.

[§] <https://github.com/HydroWIRES-PNNL/inflow-forecast-evaluation>

[¶] <https://doi.org/10.5281/zenodo.16921728>

- Laugesen, Richard, et al. "Flexible Forecast Value Metric Suitable for a Wide Range of Decisions: Application Using Probabilistic Subseasonal Streamflow Forecasts." *Hydrology and Earth System Sciences*, vol. 27, no. 4, 2023, pp. 873–893.
- Lee, Eunkyung, et al. "Developing Resilient Reservoir System Operations: A Simulation Approach Incorporating Rule Curve." *Water Resources Management*, June 2025).
- Liu, Y., et al. "Advancing Data Assimilation in Operational Hydrologic Forecasting: Progresses, Challenges, and Emerging Opportunities." *Hydrology and Earth System Sciences*, vol. 16, no. 10 October 2012), pp. 3863–3887.
- Nash, J. E., and J. V. Sutcliffe. "River Flow Forecasting through Conceptual Models Part I — A Discussion of Principles." *Journal of Hydrology*, vol. 10, no. 3 April 1970), pp. 282–290.
- Pagano, Thomas C., et al. "Challenges of Operational River Forecasting." *Journal of Hydrometeorology*, August 2014).
- Pham, Leo Triet, et al. "Evaluation of Random Forests for Short-Term Daily Streamflow Forecasting in Rainfall- and Snowmelt-Driven Watersheds." *Hydrology and Earth System Sciences*, vol. 25, no. 6 June 2021), pp. 2997–3015.
- St-Aubin, Philippe, and Bruno Agard. "Precision and Reliability of Forecasts Performance Metrics." *Forecasting*, vol. 4, no. 4 December 2022), pp. 882–903.
- Thompson, John C., and Glenn W. Brier. "The Economic Utility of Weather Forecasts." *Monthly Weather Review*, vol. 83, no. 11, 1955, pp. 249–253.
- Yang, Guang, et al. "Sensitivity of Forecast Value in Multiobjective Reservoir Operation to Forecast Lead Time and Reservoir Characteristics." *Journal of Water Resources Planning and Management*, vol. 147, no. 6 June 2021), pp. 04021027.
- Zajac, Zuzanna, et al. "The Impact of Lake and Reservoir Parameterization on Global Streamflow Simulation." *Journal of Hydrology*, vol. 548 May 2017), pp. 552–568.
- Zhang, Rongqi, et al. "Optimization of Short-Term Hydropower Scheduling with Dynamic Reservoir Capacity Based on Improved Genetic Algorithm and Parallel Computing." *Journal of Hydrology*, vol. 636 June 2024), pp. 131238.